



DRAFT International Standard

ISO/DIS 21043-5

Forensic Sciences — Part 5: Reporting

*Science criminalistique —
Partie 5: Rapports*

ICS: 07.140

ISO/TC 272

Secretariat: **SA**

Voting begins on:
2024-05-02

Voting terminates on:
2024-07-25



Please share your feedback about
the standard. Scan the QR code
with your phone or click the link

[Customer Feedback Form](#)

the committee secretariat.

ISO/CEN PARALLEL PROCESSING

THIS DOCUMENT IS A DRAFT CIRCULATED FOR COMMENTS AND APPROVAL. IT IS THEREFORE SUBJECT TO CHANGE AND MAY NOT BE REFERRED TO AS AN INTERNATIONAL STANDARD UNTIL PUBLISHED AS SUCH.

IN ADDITION TO THEIR EVALUATION AS BEING ACCEPTABLE FOR INDUSTRIAL, TECHNOLOGICAL, COMMERCIAL AND USER PURPOSES, DRAFT INTERNATIONAL STANDARDS MAY ON OCCASION HAVE TO BE CONSIDERED IN THE LIGHT OF THEIR POTENTIAL TO BECOME STANDARDS TO WHICH REFERENCE MAY BE MADE IN NATIONAL REGULATIONS.

RECIPIENTS OF THIS DRAFT ARE INVITED TO SUBMIT, WITH THEIR COMMENTS, NOTIFICATION OF ANY RELEVANT PATENT RIGHTS OF WHICH THEY ARE AWARE AND TO PROVIDE SUPPORTING DOCUMENTATION.

ISO/DIS 21043-5:2024(en)



COPYRIGHT PROTECTED DOCUMENT

© ISO 2024

All rights reserved. Unless otherwise specified, or required in the context of its implementation, no part of this publication may be reproduced or utilized otherwise in any form or by any means, electronic or mechanical, including photocopying, or posting on the internet or an intranet, without prior written permission. Permission can be requested from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office
CP 401 • Ch. de Blandonnet 8
CH-1214 Vernier, Geneva
Phone: +41 22 749 01 11
Email: copyright@iso.org
Website: www.iso.org

Published in Switzerland

© ISO 2024 – All rights reserved

Contents

	Page
Foreword	iv
Introduction	v
1 Scope	1
2 Normative references	1
3 Terms and definitions	1
4 General requirements	1
5 Report contents	2
5.1 General.....	2
5.2 Administrative requirements.....	2
5.3 Item integrity.....	3
5.4 Significant figures and measurement uncertainty.....	3
5.5 Reporting opinions.....	3
5.5.1 General.....	3
5.5.2 Reporting probabilities.....	4
5.5.3 Reporting likelihood ratios.....	4
5.5.4 Reporting estimations.....	4
6 Case file review	4
6.1 General.....	4
6.2 Technical review.....	5
6.3 Administrative review.....	5
7 Issue and control of reports	5
7.1 Issuing reports.....	5
7.2 Supplementing, amending and withdrawing reports.....	6
7.3 Controlling reports.....	6
7.4 Communications of observations and/or opinions other than by way of a report.....	6
8 Testimony	7
8.1 General.....	7
8.2 Testimony training and evaluation.....	7
Annex A (informative) The forensic process	8
Annex B (informative) Reports and other communication of information	10

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular the different approval criteria needed for the different types of ISO documents should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights. Details of any patent rights identified during the development of the document will be in the Introduction and/or on the ISO list of patent declarations received (see www.iso.org/patents).

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation on the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT) see the following URL: www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 272, *Forensic Sciences*.

The following is a list of all parts in the ISO 21043 series:

ISO 21043-1, *Forensic Sciences – Part 1: Terms and definitions*

ISO 21043- 2, *Forensic Sciences – Part 2: Recognition, recording, collection, transport and storage of items*

ISO 21043- 3, *Forensic Sciences – Part 3: Analysis*

ISO 21043- 4, *Forensic Sciences – Part 4: Interpretation*

ISO 21043-5, *Forensic Sciences – Part 5: Reporting*

Introduction

This document is part of a series of five that includes the different components of the forensic process from scene to courtroom. This document establishes requirements to safeguard the process for reporting.

In this document, the following verbal forms are used:

- “shall” indicates a requirement;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capability.

Further details can be found in the ISO/IEC Directives, Part 2.

[Annexes A](#) and [B](#) are for information only.

The notes given provide clarification of the text, examples and guidance. They do not contain requirements.

Forensic Sciences —

Part 5: Reporting

1 Scope

This document specifies requirements and provides recommendations for reporting of forensic work. It is applicable to the reporting of forensic work performed at a scene and within a forensic facility. The requirements facilitate the preparation and issuing of reports that are accurate, clear, transparent, complete, unambiguous, impartial and suitable for their intended use. This document also includes requirements relating to the review of case records and reports, the issuance and control of reports, and requirements related to testimony.

NOTE Legal requirements take precedence over any requirements in this standard.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 21043-4, *Forensic Sciences Part 4: Interpretation*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 21043-1, *Forensic Sciences Part 1: Terms and definitions* apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <https://www.electropedia.org/>
- ISO Online browsing platform: available at <https://www.iso.org/obp>

4 General requirements

Reports shall be accurate, clear, transparent, complete, unambiguous, impartial, and suitable for their intended use.

The forensic service provider shall have a documented procedure for formally authorizing examiners to issue reports.

Examiners shall not report beyond their area of expertise.

Examiners shall not report beyond what can be based on the available information.

Known limitations of methods, procedures, observations, and opinions, should be stated clearly in the report or in an appendix attached to the report. If the limitations are not stated in the report or in an appendix to the report it shall be stated that the information on the limitations is available and shall be provided upon request.

ISO/DIS 21043-5:2024(en)

Descriptions of any deviation from documented methods or procedures shall be included in the report if the deviation could have a substantial impact on the observations or opinions. The reason for any such deviation shall be explained in the report or in an appendix attached to the report.

Any terms that are not in common usage, or for which the specialist usage differs from the common use, should be clearly defined and communicated to the customer (e.g., in a glossary appended to the report).

[Annex B](#) provides information on different types of reports.

5 Report contents

5.1 General

All reports shall include:

- a) a description of the items examined including their unique identifiers;
- b) a description of the examinations undertaken;
- c) the results of examinations, including observations (as relevant) and opinions reached;
- d) an identifier of the person(s) responsible for the content of the report.

All reports should also include or reference:

- e) what was requested;
- f) who made the request;
- g) when the request was made and/or received;
- h) date items were collected or received;
- i) identifying information for scenes examined, if applicable;
- j) method(s) used;
- k) additional information relied upon during the examination (e.g., photographs, other forensic reports, task-relevant circumstances of the case that were considered);
- l) a list of the items accepted for analysis but not analysed;
- m) details of the environmental conditions under which the examinations were conducted where such conditions lie outside the expected operating range and could have affected the results of the examination;
- n) if analysis has been performed despite the suitability of the item for analysis having been questioned by the forensic service provider;
- o) a signature or e-signature of the person(s) responsible for the content of the report.

5.2 Administrative requirements

All reports shall include:

- a) a title that clearly designates the document as a report (e.g., by including the word "report" in the title);
- b) a means of uniquely identifying the report;
- c) the date of issue;
- d) a unique case identifier (e.g., case number);

ISO/DIS 21043-5:2024(en)

- e) the name of the forensic service provider;
- f) pagination (e.g., page 1 of 10);
- g) a means of signifying the end of the report;
- h) reference to any appendices or attachments.

NOTE to a): Reports are sometimes referred to as statements, certificates, or expert reports.

5.3 Item integrity

When there is an indication that the integrity of an item has been compromised prior to analysis, the description of the item in the report shall include details of the noticed irregularities such as the packaging and condition of the item as received.

The report shall also describe any mitigating actions taken.

5.4 Significant figures and measurement uncertainty

When quantitative observations are reported, the number of significant figures used to report the final values shall be supported by the measurement uncertainty of the method used.

Measurement uncertainty shall be reported where relevant and applicable.

5.5 Reporting opinions

5.5.1 General

If opinions are reported, the opinions shall conform to ISO 21043-4.

The report shall differentiate opinions from observations.

Reports that contain opinions shall include:

- a) the questions to be answered;
- b) the set of propositions, where applicable;
- c) a description of assumptions that are relevant for forming the opinion;
- d) the basis of the opinion.

If reports contain opinions and an opinion scale is used, a reproduction of the entire scale shall be included or referenced in the report.

Reports that contain opinions should include or reference:

- e) a description of any data used;
- f) a description of any statistical models used.

Opinions shall not be presented in such a way as to overstate or understate their strength.

Where the strength of the opinion is reported as a likelihood ratio, a bound of the likelihood ratio closer to the neutral value of 1 may be reported, making clear that the likelihood ratio exceeds that bound. The forensic service provider shall have a documented rationale for choosing that bound.

EXAMPLE 1 In a DNA profile comparison, an opinion could state that “the likelihood ratio is greater than 1 billion”. This would not be understating the strength of the opinion.

ISO/DIS 21043-5:2024(en)

NOTE Lowering a likelihood ratio based on limitations of the quality of the item, measurement or data, is not understating the strength of the opinion, but is following the requirements to take limitations into account in the interpretation, refer to ISO 21043-4.

Opinions, including categorical opinions, shall not be presented as if they were statements of fact.

When reporting opinions based on questions of source, the examiner should explicitly state that only the source of the item was considered, and not the activity through which the item was transferred.

5.5.2 Reporting probabilities

If probabilities are assigned using data, the data used should be described and/or referenced in the report.

If a method used for assigning probabilities produces a numerical value, the opinion should be expressed using that numerical value.

If probabilities are assigned using the examiner's judgment, the basis for that judgment should be described in the report.

5.5.3 Reporting likelihood ratios

When reporting a likelihood ratio the examiner:

- a) shall express the likelihood ratio in a way that makes it clear that it is the ratio of the probabilities of the observations given the competing propositions; and
- b) shall not transpose the conditional.

NOTE The ratio of the probabilities of the propositions, given the observations, is known as the "posterior odds". The likelihood ratio is not the same as the posterior odds. Confusing a likelihood ratio with the posterior odds is known as the fallacy of the transposed conditional, also referred to as the "prosecutor's fallacy".

The likelihood ratio may be expressed quantitatively and/or qualitatively.

If the interpretation method calculated a quantitative likelihood-ratio value, that quantitative value should be reported.

5.5.4 Reporting estimations

Estimations shall be reported numerically and should be accompanied by an expression of the uncertainty of the estimate.

6 Case file review

6.1 General

When undertaken, case file review should be performed prior to the issue of the report to the customer.

Case file review consists of one or both of the following:

- a) technical review,
- b) administrative review.

The technical and administrative reviews may be performed separately or as one process.

When undertaken, case file review may be limited to the part of the case file related to the examiner and their area of expertise:

ISO/DIS 21043-5:2024(en)

Within that part of the case file, case file review shall include an assessment of records related to recognition, recording and collecting of items, analysis of items, interpretation of observations, and the reporting of observations and opinions made by the forensic service provider.

The type of review, parts of the case file reviewed, outcome of the review, date completed, and the identity of the reviewer shall be recorded, and the records shall be retained.

6.2 Technical review

Forensic service providers shall determine when to perform technical reviews based on risk assessment and shall have documented procedures for technical reviews including:

- a) the type and proportion of case files that require technical review;
- b) how technical reviews of case files are performed;
- c) the process to be followed when examiner and reviewer disagree regarding the analysis performed, any interpretation, or reported observations or opinions.

Reports that include opinions directly based on human judgement present a higher risk. This higher risk shall be considered in the risk assessment.

Records of the risk assessments shall be retained.

NOTE 1 The risk-based decision to determine the type and proportion of cases requiring a technical review can include factors such as the type of results being reported, the potential for an error, the impact of the types of reporting errors and experience of the examiner. The proportion of cases that require a technical review can be different within and between disciplines.

Technical review(s) shall be performed by an examiner who is competent to review in the relevant area of expertise, but shall not be performed by the examiner who carried out the examination.

When technical review is required by the procedure but is intentionally not performed (e.g., as a report is required urgently by the customer), the report shall contain a disclaimer stating that the review was not completed and why it was not completed. In these circumstances, the preliminary report should be followed by a final report that has been subject to a technical review, indicating any changes as an outcome of the review.

Disagreement on opinions between a reviewer and examiner that cannot be resolved shall be recorded and retained and should be included in the report if the report includes the disputed opinion.

NOTE 2 Information about the disagreement may be limited to the factual information that there is disagreement without further explanation.

6.3 Administrative review

Forensic service providers shall determine when to perform administrative reviews based on risk assessment and shall have documented procedures for administrative review.

Administrative review(s) shall be performed by a person who is familiar with the forensic service provider's operational requirements, and who has been deemed competent to carry out the review. The review shall not be performed by the examiner who carried out the examination.

7 Issue and control of reports

7.1 Issuing reports

Forensic service providers shall have documented procedures for the issuing of reports. The procedures shall address the following:

- a) the general circumstances under which reports are issued or not issued;

ISO/DIS 21043-5:2024(en)

- b) the types of reports that may be issued, including:
 - i) the circumstances under which each type is issued;
 - ii) what competence is required to issue reports of each type;
 - iii) the circumstances when a caveat is required (e.g., if a report is released prior to technical review);
 - iv) the required format and content;
- c) which reports are to be reviewed prior to issue.

If a request is accepted, a report should be issued.

7.2 Supplementing, amending and withdrawing reports

Forensic service providers shall have documented procedures for issuing supplementary or amended reports, and for withdrawal of reports. The procedures shall include the following requirements:

- a) customers shall be notified of a supplementary, amended or withdrawn report, and the reasons for it;
- b) supplementary reports shall reference the report that was supplemented,
- c) Amended reports shall clearly identify what has changed or has been added to the report;
- d) reports shall have a unique identifier that distinguishes between reports that are withdrawn and those that are replaced.

Customers shall be provided with a copy of any supplementary or amended report.

7.3 Controlling reports

Copies of all reports, including amended, supplementary and withdrawn, shall be retained as part of the case file.

The issued report shall be traceable so that any alterations can be identified by comparing to retained copies.

All issued reports shall undergo record control.

When reports are transmitted electronically, they shall be issued in a format so as to minimize the risk of unauthorized alteration.

7.4 Communications of observations and/or opinions other than by way of a report

In the course of an investigation, it can be necessary to provide observations and/or opinions to the customer in a timely manner. This information may be provided in a format agreed upon between the customer and the forensic service provider. When provided, the information shall meet the following requirements:

- a) opinions or observations that would require a review as described in [6.2](#) shall either:
 - i) be reviewed prior to release or
 - ii) accompanied by a caveat indicating that review has not been conducted;
- b) substantial limitations of the observations and/or opinions and their preliminary nature shall be stated;
- c) a record of the communication shall be retained and included in the case file and shall include:
 - i) a copy or summary of the information provided and a reference to the case,
 - ii) the date the information was provided,

- iii) sufficient detail to determine how the information was provided, who provided the information and who the information was provided to.

In addition, observations should be checked before they are communicated.

These requirements apply irrespective of whether the information is communicated orally or in writing.

When the information is intended to be used for court, a report meeting all requirements of this standard should be issued to the customer.

8 Testimony

8.1 General

Testimony shall be accurate, clearly stated, responsive to the questions asked, impartial, and within the examiner's area of expertise.

Opinions shall not be presented in such a way as to overstate or understate their strength.

Opinions, including categorical opinions, shall not be presented as if they were statements of fact.

The expert witness should use terminology in common usage. If the usage differs from the common use, it should be clearly defined and communicated to the legal forum.

Opinions should be clearly distinguished from observations.

Observations and opinions should be supported by reports, case records, data, and/or documents.

8.2 Testimony training and evaluation

If personnel provide testimony, the forensic service provider shall have a documented procedure for testimony training, and shall have a documented procedure for assessment of competence of personnel to provide testimony.

If a testimony review is to be performed, the procedure should include:

- a) the frequency of review;
- b) who is permitted to perform the review;
- c) how the review is recorded, including identity of the reviewer and the date(s) performed;
- d) the process whereby feedback is given to the personnel providing testimony.

The assessment of competence to provide testimony should assess whether:

- e) the strength of the opinion was not overstated or understated;
- f) the basis for opinions was clearly described;
- g) clarifications and qualifications were provided (e.g., when affirmative or negative answers are requested, but the answer may be misleading);
- h) the testimony was within the area of expertise of the personnel providing testimony;
- i) the personnel providing testimony used terminology that was appropriate to the legal forum;
- j) the testimony was presented in an impartial manner, not misleading, and responsive to the questions asked.

Annex A (informative)

The forensic process

In [Figure A.1](#) below is a general scheme intended only to illustrate how the parts of the ISO 21043 series are related.

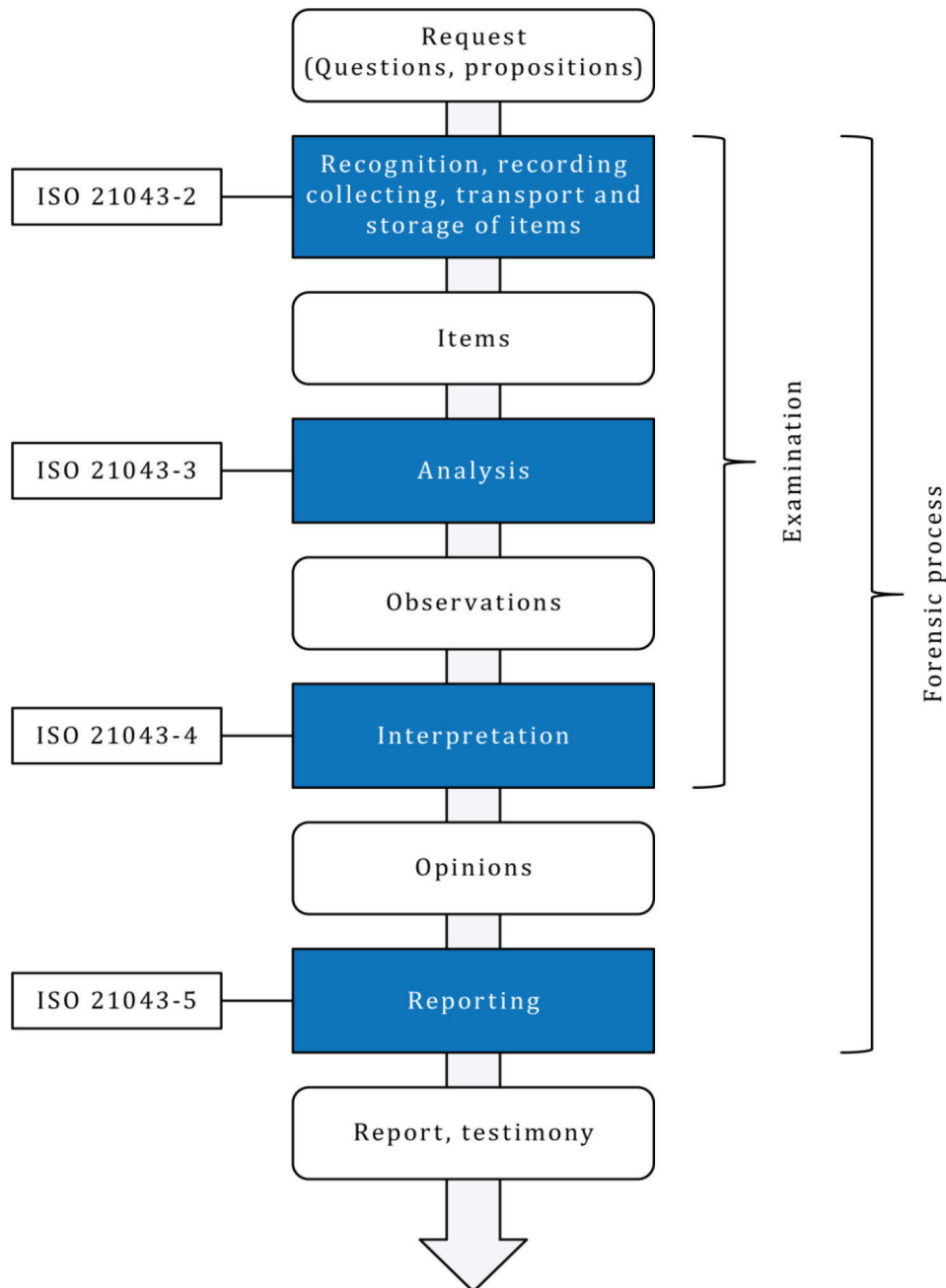


Figure A.1 — Forensic Process

The blue boxes indicate processes and are linked to the various parts of the standard (indicated with labels on the left side). The white boxes with rounded corners can be the input or output of a process.

The forensic process can be very diverse, and the arrow only illustrates the general flow. In a particular forensic process some actions might be skipped or repeated.

Annex B **(informative)**

Reports and other communication of information

This standard distinguishes reports and other communications of information.

Reports are designated as such in the title (e.g., by including the word "report" in the title) and are compliant with the requirements defined in this standard.

Reports can vary in three different, independent aspects:

a) *The content and purpose of the report*

Some reports such as investigative or evaluative reports, contain opinions. Some reports only include observations. Some reports include both observations and opinions. Reports can also be both investigative and evaluative. Reports can include interpretations of observations by providing explanations for the observations or by providing an estimation based on observations. A single report can serve multiple purposes.

b) *The position of the report in an administrative sequence*

Reports can be preliminary, amended, supplemented, withdrawn, or final.

Reports can be a summary of preliminary results or a summary of final results that is provided to customers prior to the production of a final report.

Amended reports are issued when a technical or administrative error is identified, or if a previously issued report needs to be changed. An amended report replaces the original report.

Supplemented reports are issued after additional analysis, or to add information which was not included in the original report.

c) *The channel of delivery of the report*

Reports can be delivered on paper or digitally.

Communications of information other than by way of a report include digital messages, such as e-mails, and oral communication.

E-mails can be used as a channel for issuing a report, but also for digital messages. Digital messages can be used, for example, if the result of a particular test, search or examination was negative, leading to an end of a particular line of enquiry, or a potential candidate was found in a database and further material might be required for analysis.

Oral communication (apart from oral testimony) can be in person, over the phone, or by video conference or other electronic means. Such communications are intended for information and guidance during the course of an investigation or when an examination has yielded negative results leading to the end of an inquiry.